Regression

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Outlines

- Supervised Learning (recap)
- Linear Regression
- Handle Categorical Variables
- Feature Selection
- Regression Performance

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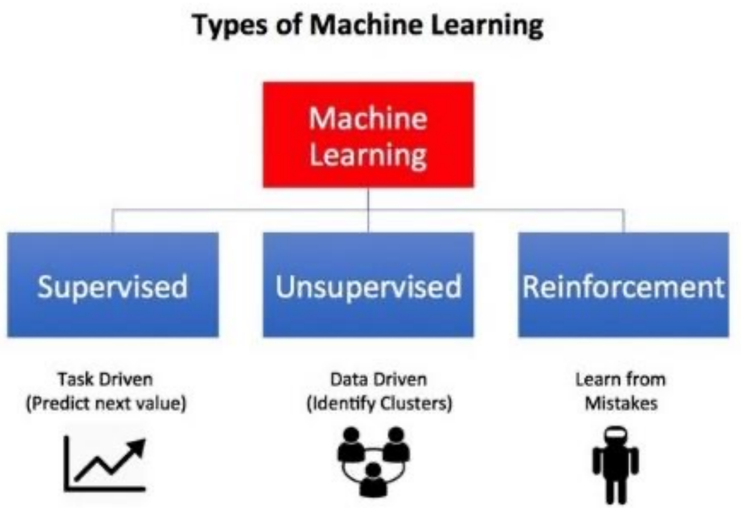
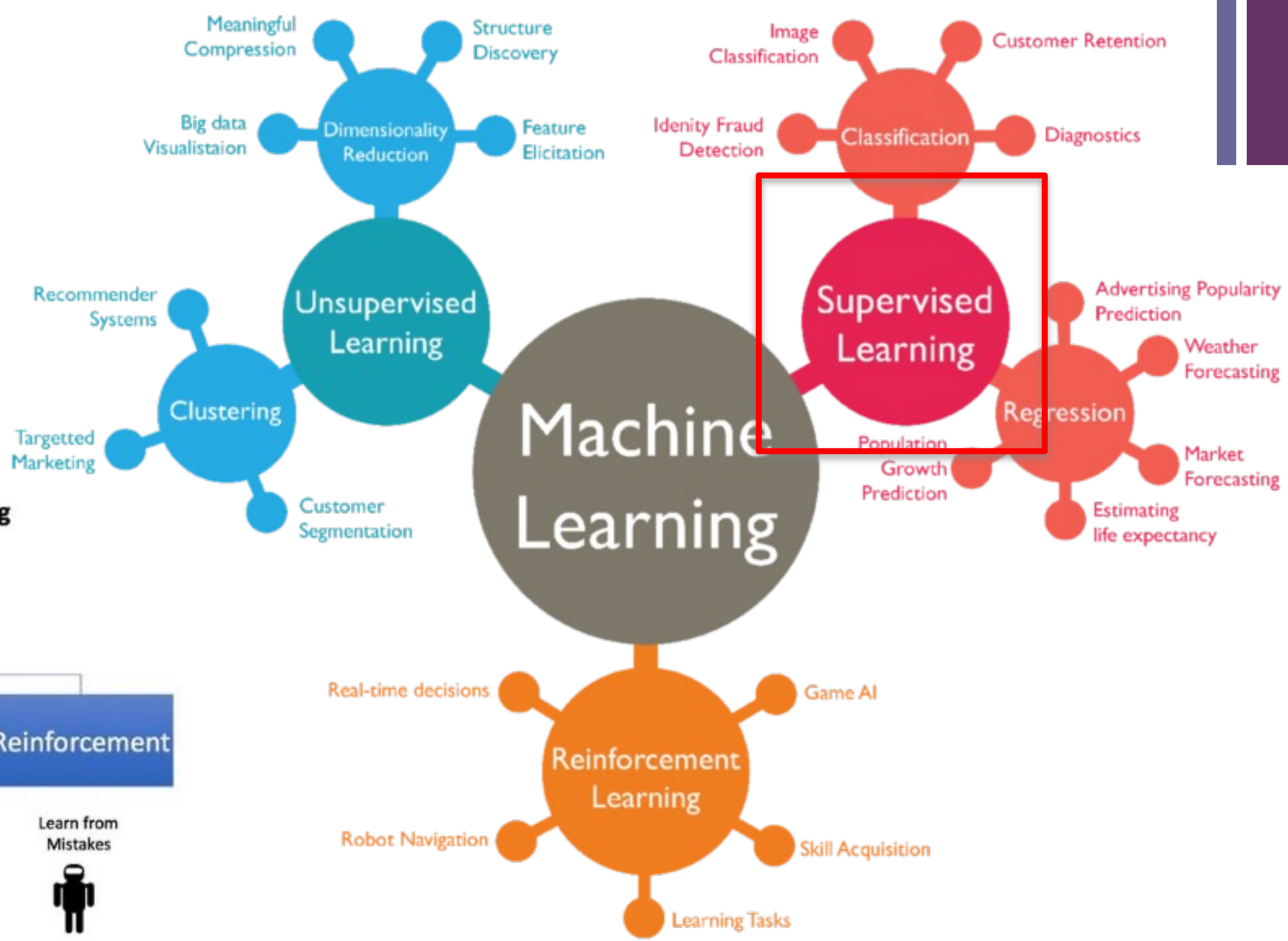
Supervised Learning (recap)

+ Machine Learning



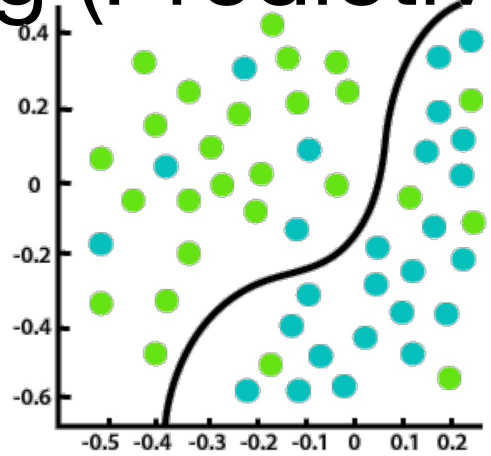


Supervised Learning (Predictive Task)

inputs				target
Age	**Temp**	**Gender**	**Smell**	**Covid**
25	39.0	Female	No	Yes
35	38.9	Female	No	Yes
32	36.5	Male	Yes	No

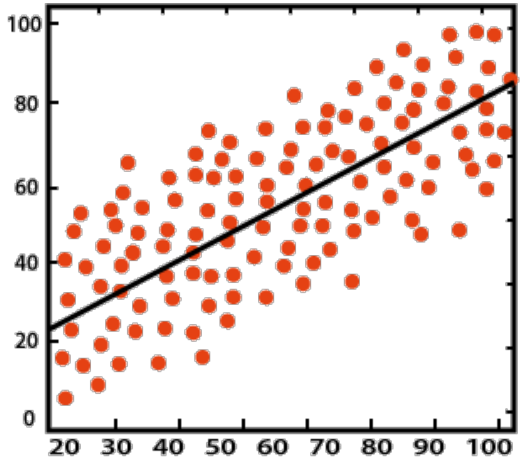
- Goal: To learn a prediction model mapping from inputs to output.

- Data without label (answer) is meaningless!
- Label should be provided by experts!



Classification

- Target is categorical variable.
- <u>Example</u>
- Covid diagnosis (yes/no)
- Disease diagnosis from gait information:
  - 1) Normal,
  - 2) Sick/Knee OA
  - 3) Sick/Parkinson



Regression

- Target is numeric variable.
- <u>Example</u>
- PD's state diagnosis from movement data.
- Glucose level prediction from breath particles.

There are two main processes: Train/Test

1) Training Phase: Model Construction

Training Data

inputs

target

Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	Yes
35	50,000	Female	Nontaburi	Yes
32	35,000	Male	Bangkok	No

2) Testing Phase: Model Evaluation, Model Assessment
Also called "prediction, inference, scoring"

Testing Data

Age	Income	Gender	Province	Purchase
25	25,000	Female	Bangkok	?

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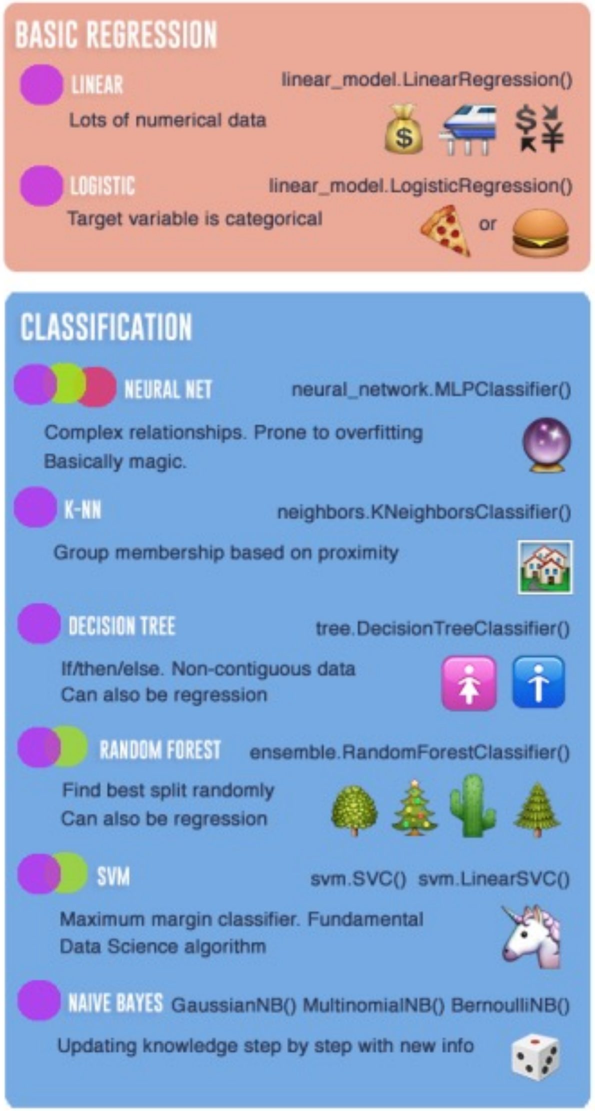
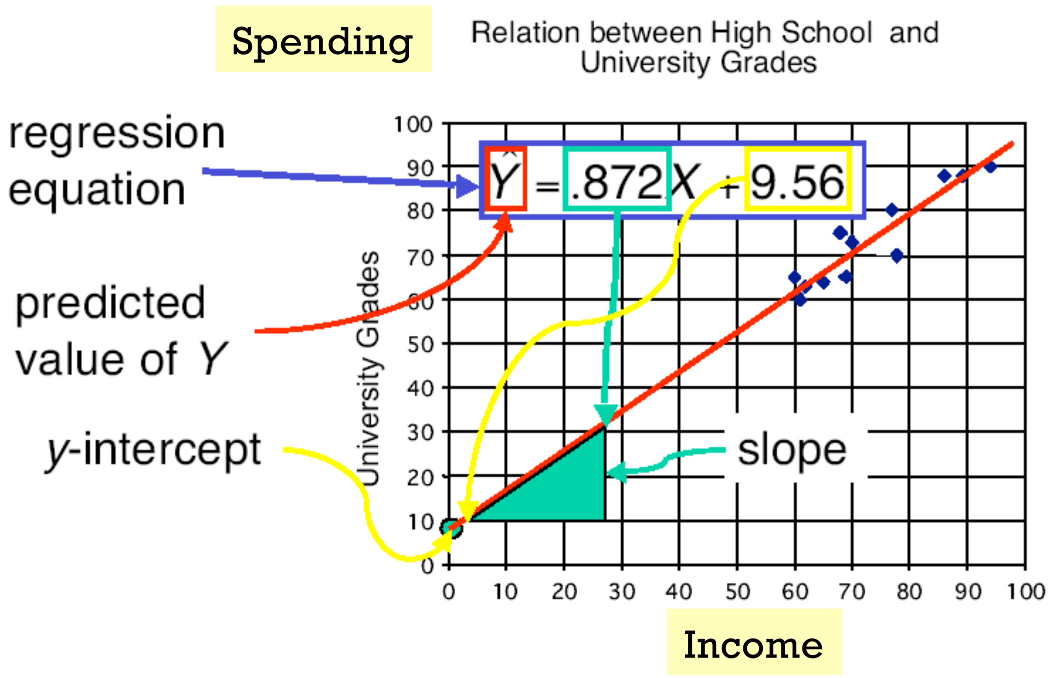
Prediction Algorithms

- Decision Tree
- (Logistic) Regression
- Neural Networks (NN)
- kNN
- Support Vector Machine
- Deep Learning



Linear Regression

Linear Regression



weight, coefficient

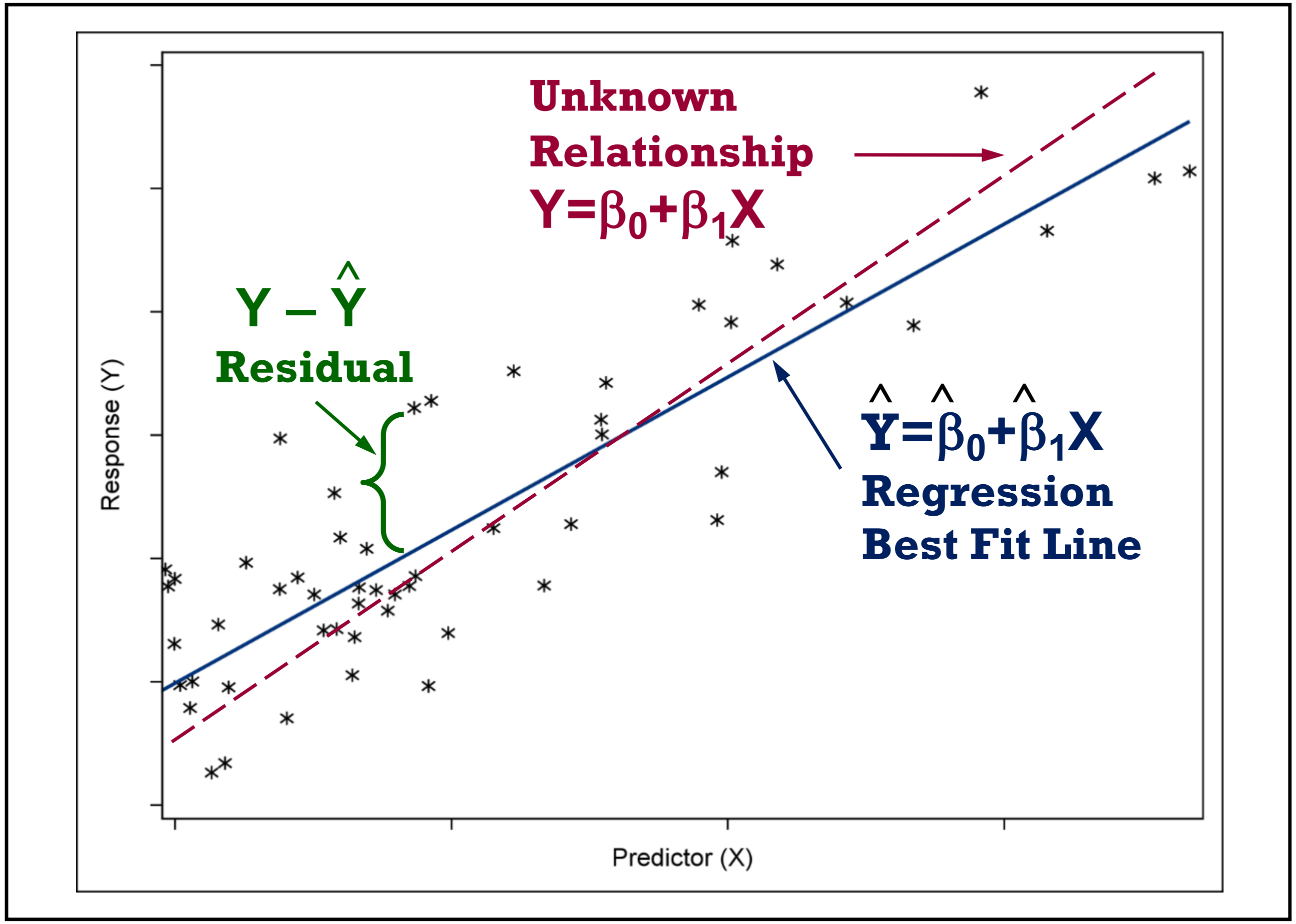
$$\hat{y} = \widehat{w}_0 + \widehat{w}_1 x_1 + \widehat{w}_2 x_2$$

target intercept input

- The least square method aims to minimize the following term

$$\sum_{training\ data} (y_i - \hat{y}_i)^2$$

Ordinary Least Squares (OLS) Regression



Multiple Linear Regression Model Matrix Multiplication Approach

	inputs	target
Age	**Income in k$**	**Spending**
25	25	400
35	50	500
32	35	550

$$Y=\beta_0(1)+\beta_1X_1+\beta_2X_2+\varepsilon$$

$$[Y] = [X][\beta]$$
$$[\beta]=[X]^{-1}[Y]$$

General least-squares solution:

$\boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$

$$\mathbf{y} = \begin{bmatrix} 400 \\ 500 \\ 550 \end{bmatrix} \qquad \mathbf{X} = \begin{bmatrix} 1 & 25 & 25 \\ 1 & 35 & 50 \\ 1 & 32 & 35 \end{bmatrix} \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

In this toy example, $\mathbf{X}$ is 3×3 and invertible, so it fits **exactly** and you can also use:

$\boldsymbol{\beta} = \mathbf{X}^{-1}\mathbf{y}$

$$\boldsymbol{\beta} = \begin{bmatrix} -250 \\ \frac{110}{3} \\ -\frac{32}{3} \end{bmatrix} \approx \begin{bmatrix} -250 \\ 36.6667 \\ -10.6667 \end{bmatrix}$$

$$\widehat{Spending} = -250 + 36.6667(\text{Age}) - 10.6667(\text{Income in k\$})$$

https://sites.google.com/site/mystatistics01/regression-correlation-analysis/multiple-linear-regression-step

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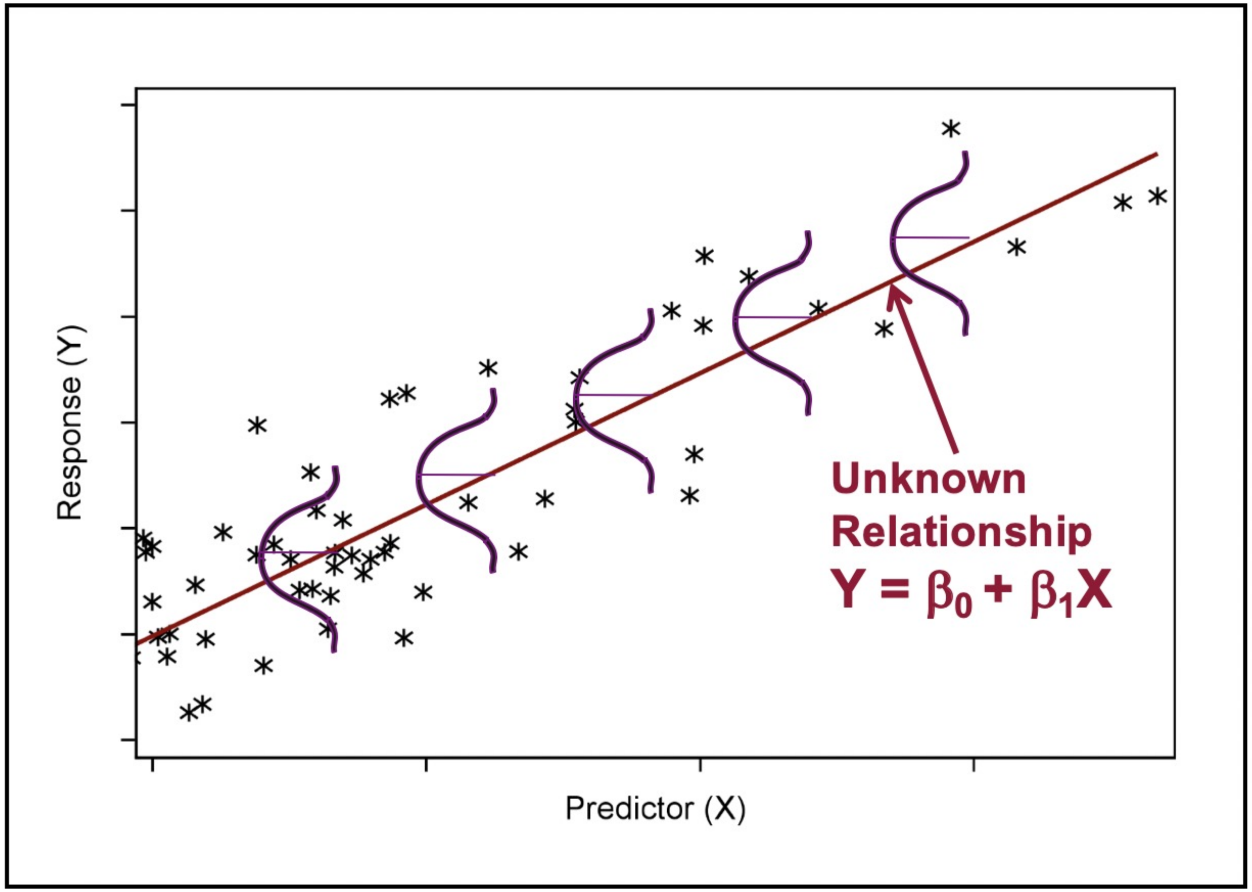
Linear Regression Assumption

Spending = 500 + 10*Income10K + 2*Age



- Linear relationship between (y, x_i). [Pearson correlation]
- Error is normal distributed. [remove outliers, log-transformation]
- Error has equal variance (homoscedasticity) [remove outliers, log-transformation]
- Errors are independent from each other. [design new data correction]

Linear Regression Limitation

- Manage missing value
- Handle outliers (skewness)
- Handle categorical variables (dummy codes)
- Variable selection:
  - Wrapper (all combinations)
  - Forward, Backward, Stepwise
- Accounted for nonlinearities

Remarks:
- Require good data preparation
- Variable selection

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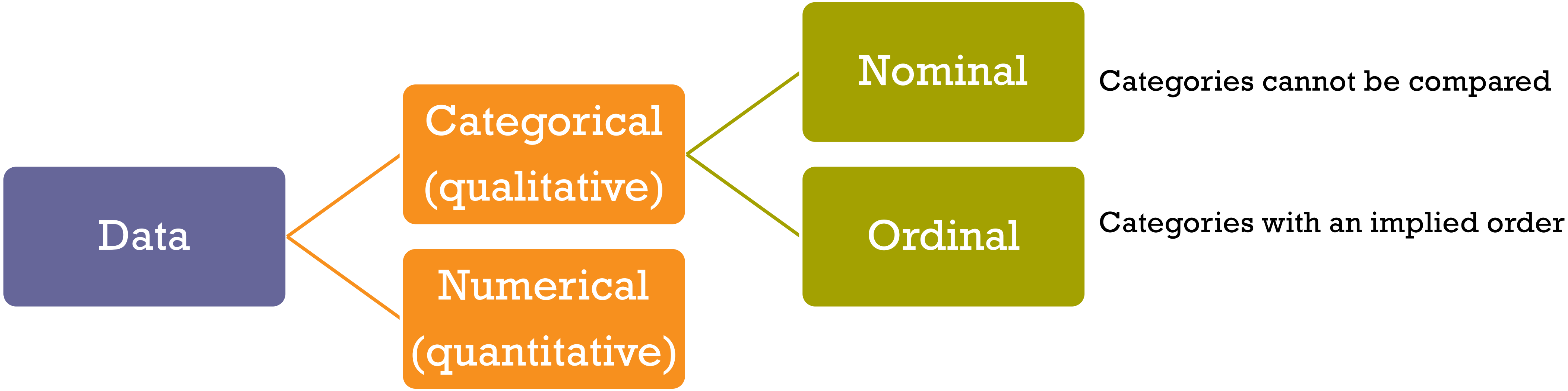
Handle Categorical Variables

Dummy coding, OneHotEncoder

+ Terminology: Kinds of data (recap)



Ordinal: Recode

Grade	GradeNum
A	4.00
B+	3.50
B	3.00
C+	2.50
C	2.00
D+	1.50
D	1.00
F	0.00

Size	SizeNum
XL	5
L	4
M	3
S	2
XS	1

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+ Categorical

- Dummy coding = (n-1) dummy codes

Branch	BranchNum	D_BKK	B_Patum	D_Non	D_BKK	B_Patum	
BKK	1	1	0	0	1	0	
Patumtani	2	0	1	0	0	1	
Nontaburi	3	0	0	1	0	0	reference

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Feature Selection

Feature Selection Approach

- Sequetial Feature Selection:
  - Sequential Feature Selection
  - Backward Feature Elimination
- Best Subet Selection (create all combinations & pick the best one)

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Sequential Feature Selection

Start with no features, then add the feature that improves performance the most at each step.

Goal: Predict *Spending*

Candidate features:

```
Age, Income, Education, Gender
```

Step 0: No features

- Model: intercept only
- RMSE = 120

Step 1: Try adding one feature at a time

Feature added	RMSE
Age	80
Income	**60** ✅
Education	95
Gender	110

➡️ Select **Income**

Step 2: Try adding one more feature (given Income)

Features	RMSE
Income + Age	**45** ✅
Income + Education	58
Income + Gender	62

➡️ Add **Age**

Step 3: Try adding another feature

Features	RMSE
Income + Age + Education	46
Income + Age + Gender	47

➡️ No improvement → **STOP**

✅ Final selected features

```code
Income, Age
```

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Backward Feature Elimination

Backward elimination starts with all features and iteratively removes the least useful one based on performance impact.

Goal: Predict *Spending*

Candidate features:

```
Age, Income, Education, Gender
```

Step 0 — Start with ALL features

```code
Age, Income, Education, Gender
```

Baseline: **RMSE = 60**

Step 1 — Try removing ONE feature at a time

Feature removed	Remaining features	Val RMSE
Age	Income, Education, Gender	95 ❌
Income	Age, Education, Gender	110 ❌
Education	Age, Income, Gender	62 ❌
Gender	**Age, Income, Education**	**58** ✅ (better)

✅ **Decision:** remove **Gender** (RMSE improves: 60 → 58)

Step 2 — Current set

```code
Age, Income, Education
```

Baseline: **RMSE = 58**

Try removing one:

Feature removed	Remaining features	Val RMSE
Age	Income, Education	88 ❌
Income	Age, Education	105 ❌
Education	**Age, Income**	**57** ✅ (better)

✅ **Decision:** remove **Education** (58 → 57)

Step 3 — Current set

```code
Age, Income
```

Baseline: **RMSE = 57**

Try removing one:

Feature removed	Remaining features	Val RMSE
Age	Income	80 ❌
Income	Age	92 ❌

🔴 **Decision:** STOP

✅ Final selected features

```code
Income, Age
```

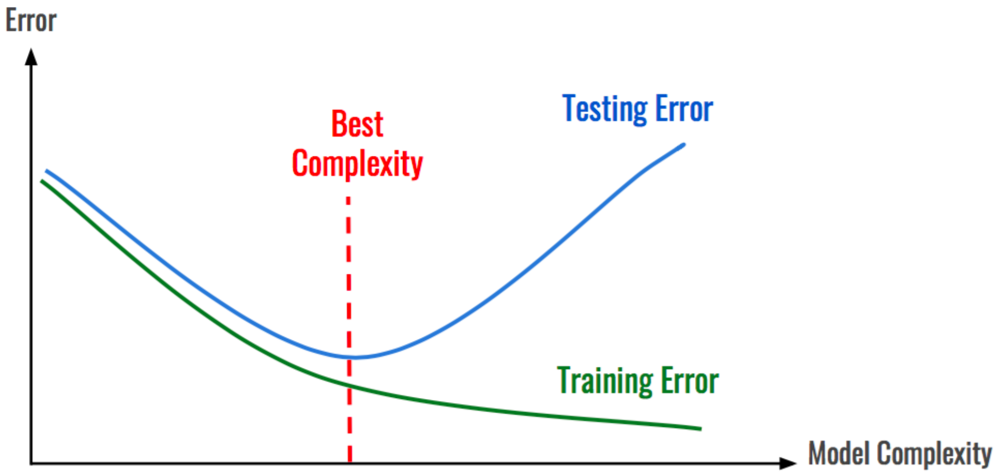
Evaluation (Train/Test Split)

Training Data

Age	Income	Purchase
25	25,000	Yes
35	50,000	Yes
32	35,000	No

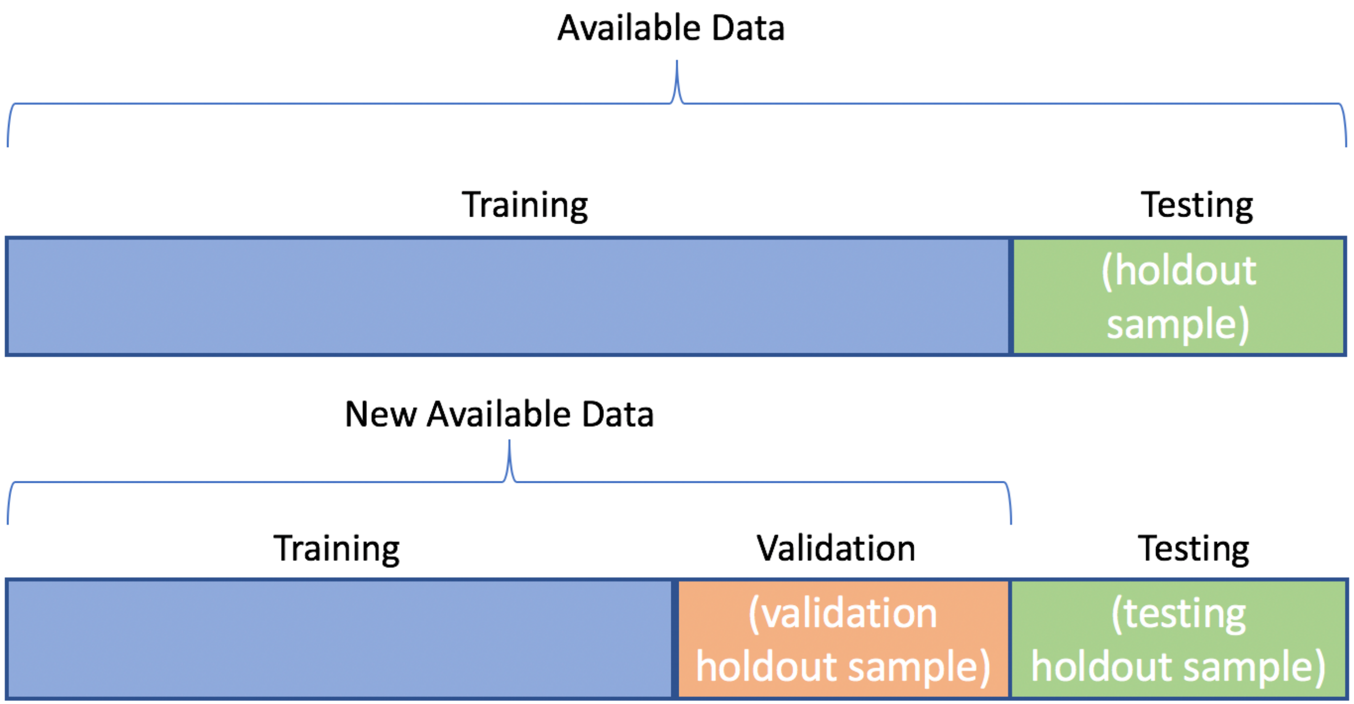
Testing Data

Age	Income	Purchase
27	35,000	Yes
23	20,000	No
45	34,000	No



+ Train (Validation) & Test

- Training data = Textbook
- Validation data = Exercise
- Testing data = Final exam



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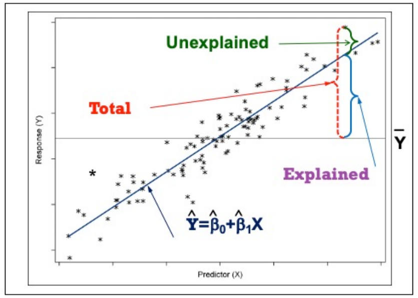
Regression Performance

Regression	
'mean_absolute_error'	`sklearn.metrics.mean_absolute_error`
'mean_squared_error'	`sklearn.metrics.mean_squared_error`
'r2'	`sklearn.metrics.r2_score`

Coefficient of Determination

Regression	
‘mean_absolute_error’	`sklearn.metrics.mean_absolute_error`
‘mean_squared_error’	`sklearn.metrics.mean_squared_error`
‘r2’	`sklearn.metrics.r2_score`



$$R^2 = \frac{SSM}{SST} = 1 - \frac{SSE}{SST}$$

id	chol (x)	bp (y)	predict	error	squred error (SE)	guess	(y - y_bar)	squred total (ST)
1	437	194	196.1897	(2.1897)	4.7948	143.4286	50.5714	2,557.4694
2	264	121	141.4179	(20.4179)	416.8906	143.4286	(22.4286)	503.0408
3	249	131	136.6689	(5.6689)	32.1364	143.4286	(12.4286)	154.4694
4	297	159	151.8657	7.1343	50.8982	143.4286	15.5714	242.4694
5	243	123	134.7693	(11.7693)	138.5164	143.4286	(20.4286)	417.3265
6	272	161	143.9507	17.0493	290.6786	143.4286	17.5714	308.7551
7	161	115	108.8081	6.1919	38.3396	143.4286	(28.4286)	808.1837
average	274.7143	143.4286		SSE	972.2548		SST	4,991.7143
				MSE	138.8935			
				RMSE	**11.7853**			
	R^2	**1 - (SSE/SST)**	**0.8052**					

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Thank you & any questions